

EEE308L

Electronic Circuits II Laboratory

Lab Report

Section:06



Inspiring Excellence

Group No:04

Experiment no:03

Name of the experiment: Study of Operational Amplifier as Precision Rectifier

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Experiment Name: study of Operational Amplifier as Precision Rectifier.

Objective: To study op amp as precision rectifier.

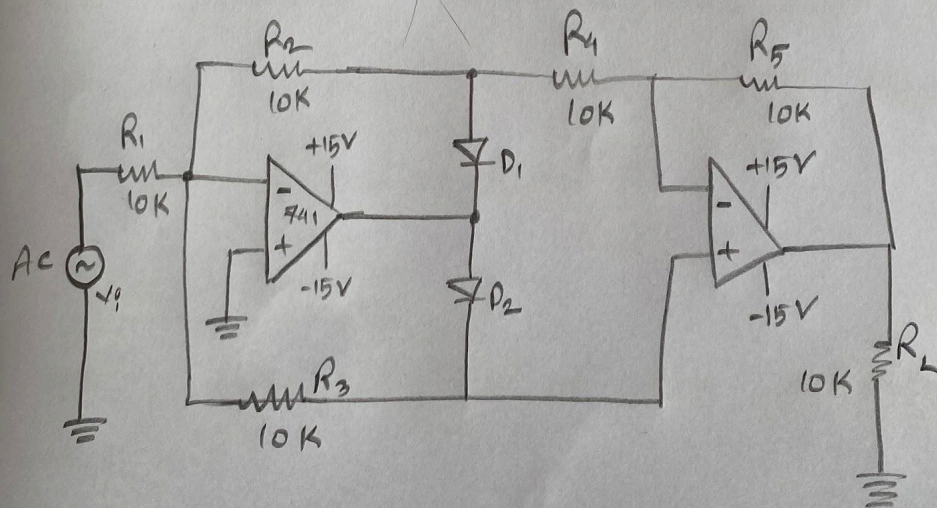
Theoretical Background:

A precision rectifier is a circuit that uses an operational amplifier and diodes to rectify small AC signals accurately. Unlike a normal diode rectifier, it eliminates the diode forward voltage drop problem (about $0.7V$), allowing even very small signals to be rectified. In this experiment, an op-amp (741) with diodes and resistors is used to create a precision rectifier circuit that converts the input AC signal into a rectified output signal with higher accuracy.

Equipments:

- ① IC: 741 - 2 pcs
- ② Resistor: 10K - 6 pcs
- ③ Diode: 1N4148 - 2 pcs
- ④ AT-7000 - 1 Unit
- ⑤ DC power supply - 1 Unit
- ⑥ AC power supply - 1 Unit
- ⑦ Oscilloscope - 1 unit
- ⑧ Multimeter - 1 unit
- ⑨ Wires

Circuit Diagram:



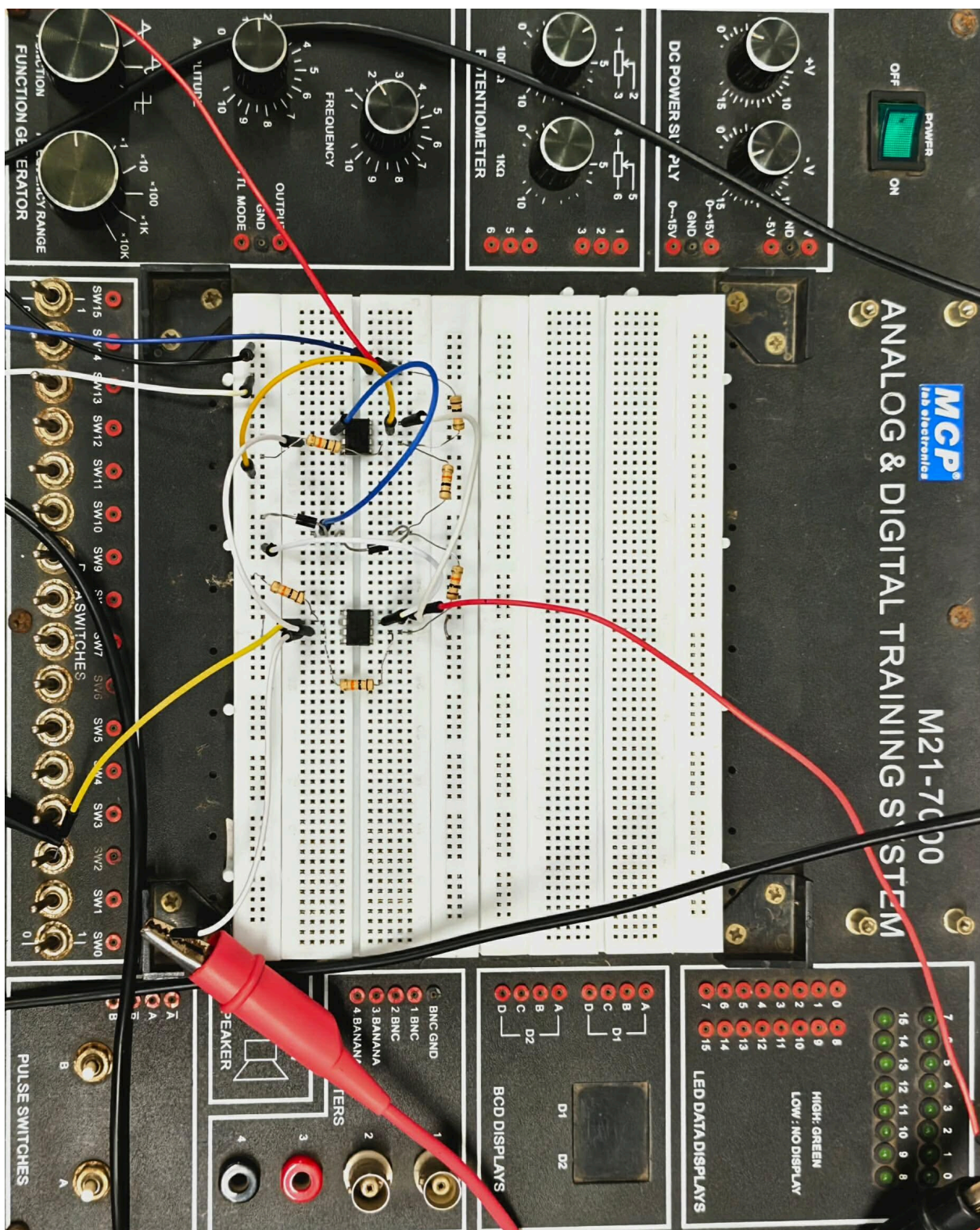


Fig: Circuit Setup

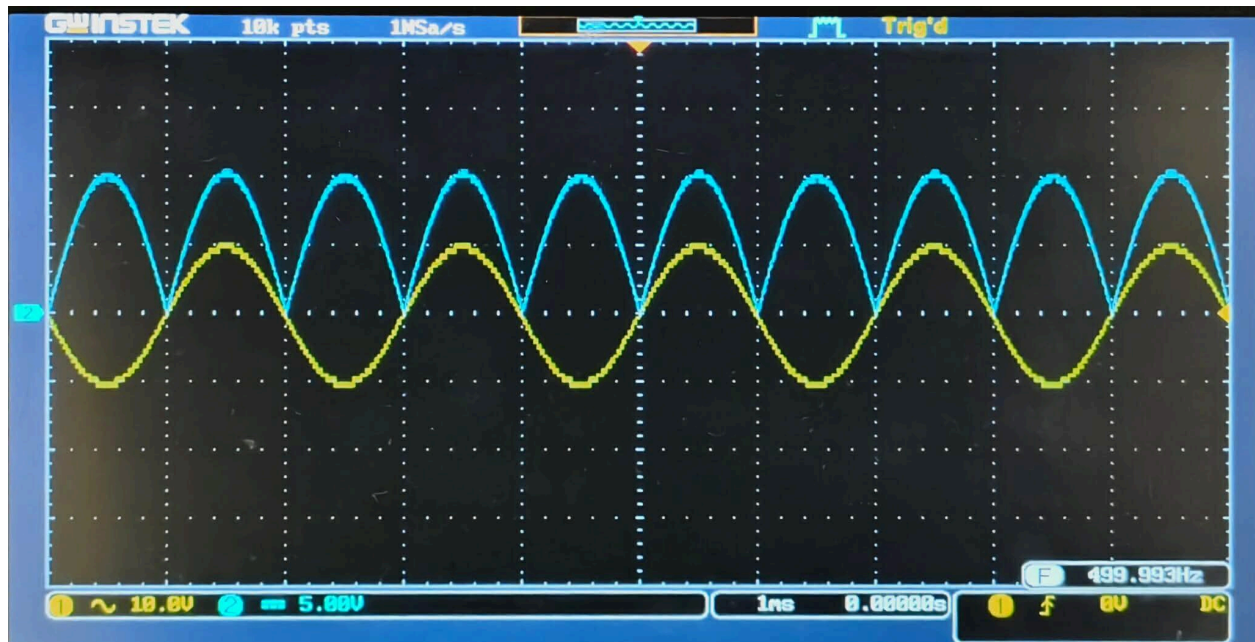


Fig: Rectifier

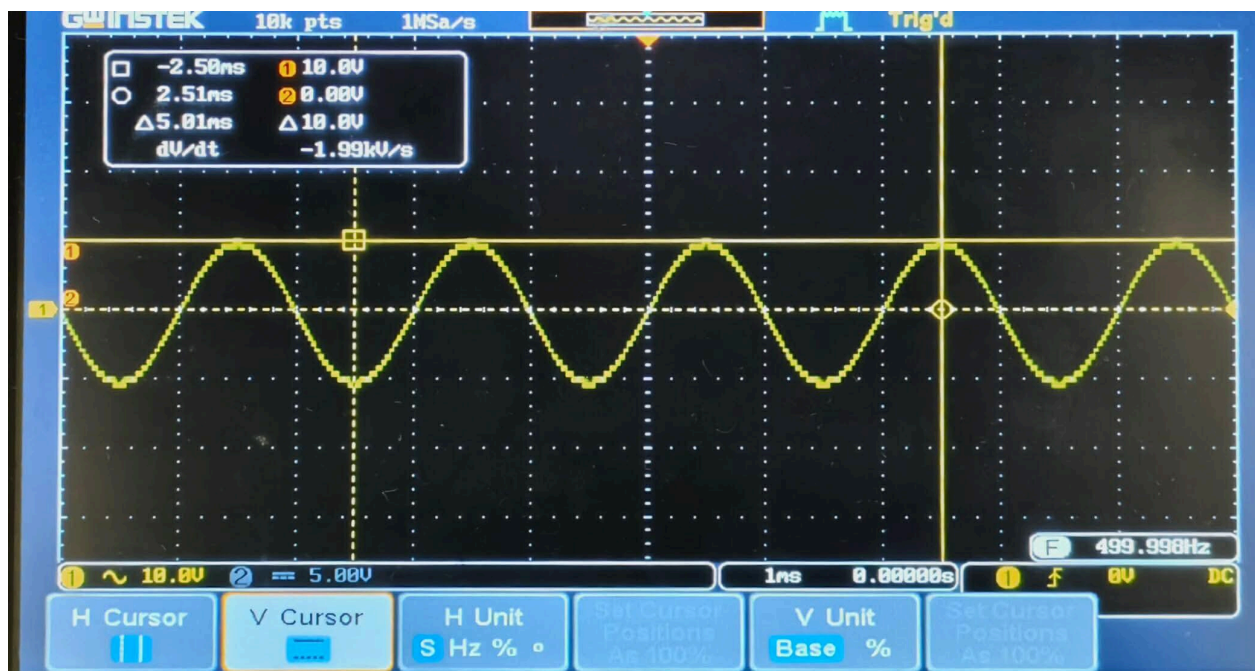


Fig: peak to peak of input

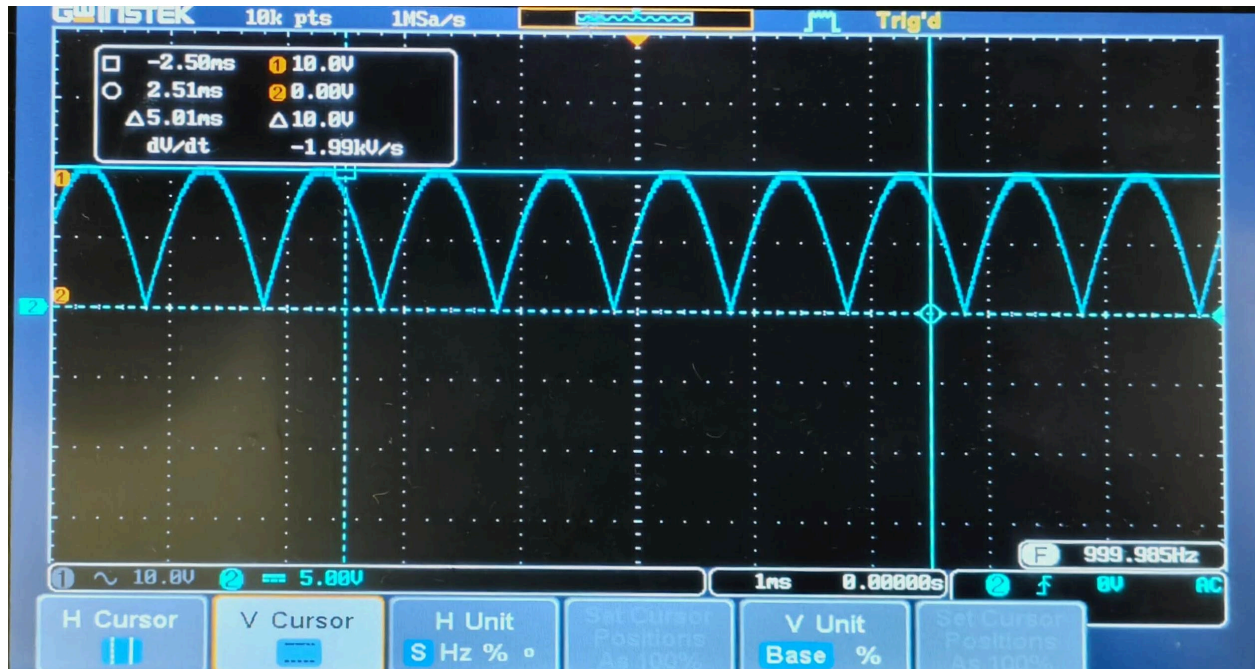


Fig: peak to peak of output

Discussion: In this experiment, a precision rectifier circuit was built using an operational amplifier, resistors and diodes.

The input signal from the function generator was applied to the circuit and the output waveform was observed using an oscilloscope. The op-amp amplified the input signal and controlled the diode operation so that the circuit could rectify even small voltage signals. The observed output waveform confirmed that the circuit successfully rectified the input signal with minimal distortion.

Question/Answers:

① Explain the operation of the precision rectifier?

Ans: A precision rectifier uses an operational

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amplifiers and diodes to convert an AC input signal into a rectified output signal. When the input signal is positive, the op-amp drives the diode into conduction and the output follows the positive part of the input signal. When the input signal is negative, the diode becomes reverse biased and the output becomes zero. It follows another path depending on the circuit type. The op-amp compensates for the diode voltage drop and allows accurate rectification even for very small signals.

② Why is it called "precision"?

Ans: It is called a precision rectifier because it can rectify very small

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input signals accurately without being affected by the diode's forward voltage drop. The operational amplifier amplifies the signal and cancels the diode threshold voltage, which allows the circuit to produce a more accurate output compared to a normal rectifier.

Conclusion 8

From this experiment, it was verified that an operational amplifier can be used to build a precision rectifier circuit. The circuit successfully rectified small input signals without the voltage drop problem of ordinary diodes. Therefore, precision rectifiers are useful in applications where accurate rectification of low-level signal is required.